

Naomi Baes

NLP & Computational Social Science Researcher

Semantic Change • Mental Health Concepts • NLP Benchmarks

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Citizenship: Australian & Romanian (EU)

Public Academic CV, last updated June 2026. For a detailed or updated CV, please contact me by email.

Professional Profile

NLP and computational social science researcher studying how word meanings change across time, registers, and communicative contexts. My work develops theory-informed methods for modeling and evaluating lexical semantic change in diachronic corpora, with applications to the mental health domain, social psychology, and socially contested language. I led the development of *SIBling* and *LSC-Eval*, frameworks for modeling and evaluating lexical semantic change using contextual embeddings, historical corpora, large language models, and synthetic benchmarks. My broader research connects NLP evaluation, psychological theory, and social science measurement.

Education

PhD	Doctor of Philosophy, NLP and Social Psychology Melbourne School of Psychological Sciences <i>Supervisors: Professor Nick Haslam and Dr Ekaterina Vylomova</i> <i>Advisors: Professor Yoshihisa Kashima and Professor Charles Kemp</i> <i>Thesis: A Multidimensional Framework for Evaluating Lexical Semantic Change with Social Science Applications</i>	University of Melbourne Expected June 2026
GradDip	Graduate Diploma in Psychology (Advanced), First Class Honours Melbourne School of Psychological Sciences <i>Thesis: The semantic inflation of “trauma” in psychology</i>	University of Melbourne 2021
GradDip	Graduate Diploma in Psychology, First Class Honours Melbourne School of Psychological Sciences	University of Melbourne 2020
BA	Bachelor of International Development Studies Faculty of Education and Arts	Australian Catholic University 2014

Technical Skills

Programming	Python, R, Bash; Git/GitHub; $\LaTeX$
NLP & ML	Hugging Face Transformers, Sentence Transformers, spaCy, NLTK, scikit-learn; contextual embeddings and transformer-based representations
Methods	Lexical semantic change analysis, diachronic WSD, benchmark construction, LLM-generated synthetic data, human-LLM evaluation, corpus preprocessing, lexicon-based measures
Statistics	Regression, logistic regression, mixed-effects models, ANOVA, <i>t</i> -tests, MDS, PCA
Research computing	HPC/SLURM workflows; large-scale corpus processing

Selected Publications

Google Scholar: h-index 9 (as of June 2026).

Natural Language Processing and Computational Linguistics

- Cassotti, P*, **Baes, N.***, De Pascale, S.*, De Sá, J. M. C., Periti, F., Haslam, N., Geeraerts, D., & Tahmasebi, N. (2026). A sense-level benchmark for denotational and connotational meaning relations. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. [Accepted] [\[link\]](#)
- Baes, N.**, Merx, R., Haslam, N., Vylomova, E., & Dubossarsky, H. (2025). LSC-Eval: A general framework to evaluate methods for assessing dimensions of lexical semantic change using LLM-generated synthetic data. In *Findings of the Association for Computational Linguistics: ACL 2025* (pp. 10905–10939). [\[link\]](#)
- Baes, N.**, Haslam, N., & Vylomova, E. (2024). A multidimensional framework for evaluating lexical semantic change with social science applications. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*

(Volume 1: Long Papers) (pp. 1390–1415). [\[link\]](#)

Baes, N., & Haslam, N. (2026). Threshold-calibrated word sense disambiguation: Semantic broadening without sense redistribution in schizophrenia. In *Proceedings of the 6th International Workshop on Computational Approaches to Language Change*. [\[link\]](#)

Baes, N., Haslam, N., & Vylomova, E. (2023). Semantic shifts in mental health-related concepts. In *Proceedings of the 4th Workshop on Computational Approaches to Historical Language Change* (pp. 119–128). [\[link\]](#)

Muhammad, S. H., Ousidhoum, N., Abdulmumin, I., Wahle, J. P., Ruas, T., Beloucif, M., de Kock, C., Surange, N., Teodorescu, D., [...], Babakov, N., **Baes, N.,** Bucur, A.-M., [...], Zhou, Y., & Mohammad, S. M. (2025). BRIGHTER: BRIdging the gap in human-annotated textual emotion recognition datasets for 28 languages. In *Proceedings of ACL 2025. Best Resource Paper Award*. [\[link\]](#)

Computational Social Science and Psychology Applications

Baes, N., Raszewski, L., Vylomova, E., Haslam, N., & de Kock, C. (2026). A multidimensional computational analysis of dehumanization in incel discourse. *Proceedings of the 20th International AAAI Conference on Web and Social Media*. [\[link\]](#)

Haslam, N., & **Baes, N.** (2024). What should we call mental ill health? Historical shifts in the popularity of generic terms. *PLOS Mental Health*, 1(1), e0000032. [\[link\]](#)

Baes, N., Vylomova, E., Zyphur, M., & Haslam, N. (2023). The semantic inflation of “trauma” in psychology. *Psychology of Language and Communication*, 27(1), 23–45. [\[link\]](#)

Selected Research Experience

Research Internship, Change is Key!

University of Gothenburg

Department of Philosophy, Linguistics, and Theory of Science; Supervisor: Nina Tahmasebi

Aug–Sep 2025

- Co-developed an international benchmark study of sense relations with an interdisciplinary team across computational linguistics, historical linguistics, theoretical linguistics, and psychology - accepted to ACL Main 2026.
- Contributed connotational meaning annotations, inter-annotator reliability analysis, LLM prompting experiments, error analysis, and manuscript drafting.
- Presented invited talks: University of Gothenburg, Utrecht University, and the National Research Council Canada.

Doctoral Researcher

University of Melbourne

Melbourne School of Psychological Sciences

Jun 2023–Jun 2026

- Led development of SIBling, a theory-driven framework for modeling lexical semantic change across dimensions of Sentiment, Intensity, and Breadth, grounded in linguistics, NLP, and social psychology - accepted to ACL Main 2024.
- Led development of LSC-Eval, a synthetic benchmarking framework for evaluating lexical semantic change methods using historical corpora, WordNet, and large language models - accepted to ACL Findings 2025.
- Applied these methods to psychology, media, and everyday language corpora to study how mental health concepts and socially contested psychological terms shift in meaning over time.

Research Assistant, Concept Creep Project

University of Melbourne

Melbourne School of Psychological Sciences; short-term and concurrent research contracts

Jul 2023–Jul 2025

- Engineered and optimized a reproducible NLP workflow for evaluating conceptual change in ProQuest/TDM Studio corpora, including parallel processing and GPU-backed transformer inference on Amazon SageMaker.
- Built workflows for large-scale corpus processing under data export restrictions, enabling reusable analysis of historical news corpora within the research lab.
- Identified and consulted domain experts and helped construct new text data sources, including New York Times and PubMed corpora, expanding the lab’s repository of diachronic text resources.
- Supported students in implementing SIBling-style NLP workflows for corpus-based semantic change analysis.

Predocutorial Research Assistant, Concept Creep Project

University of Melbourne

Melbourne School of Psychological Sciences; full-time fixed-term contract

Jan 2023–Jun 2023

- Standardized and implemented NLP workflows for preprocessing and large-scale corpus analysis, including cleaning, lemmatization, dependency parsing, normalized term frequencies, collocates, and part-of-speech based feature extraction.
- Analyzed large-scale corpus data using exploratory methods including MDS and PCA, and hypothesis-testing methods including OLS/GLS regression, logistic regression, ANOVA, and *t*-tests.
- Contributed to 13 concept creep projects, including projects that led to peer-reviewed publications, public-facing research outputs, and media coverage.

- Developed interdisciplinary collaborations with researchers in the Faculty of Engineering and Information Technology and the School of Computing and Information Systems.

Additional Research Roles University of Melbourne, RMIT University, and Swinburne University of Technology

Held earlier research roles in computational cognitive science, social/personality psychology, leadership research, and corpus-based studies of psychological concepts. This work contributed to peer-reviewed publications in PNAS, BJSP, PLOS ONE, Frontiers in Psychology, and Psychology of Language and Communication.

Selected Awards, Grants, and Research Support

- **AUD \$135,000** Australian Government Research Training Program Scholarship, University of Melbourne, 2023–2026.
- **AUD \$15,000** Seed Funding, Hallmark Research Initiative: Fighting Harmful Online Communication, University of Melbourne, 2025–2026. Seven-member team grant for automatic evaluation of mental health stigma in online communication, led by Yulia Otmakhova.
- **AUD ~20,000** In-kind research support, Concept Creep Grants, University of Melbourne, 2023–2025.
- **AUD ~6,500** In-kind research support, Change is Key!, University of Gothenburg, 2025.
- **AUD \$4,000** RHD Career Development Support Scheme, Melbourne School of Psychological Sciences, 2025.
- **Team** Best Resource Paper Award, Association for Computational Linguistics, 2025.

Selected Academic Activities

Talks and Presentations

- **Invited talks:** Utrecht University, University of Gothenburg, and National Research Council Canada, 2025.
- **Selected conference presentations:** ACL 2024; Findings of ACL 2025; LChange at EMNLP 2023; LChange at EACL 2026; ICWSM 2026 (presentation materials prepared for accepted paper).

Leadership and Service

- **Program Chair**, International Workshop on Computational Approaches to Language Change, EACL 2026.
- **Language Co-Lead**, BLEnD SemEval-2026 Shared Task on Evaluating Cultural Awareness in Language Models.
- **Program Committee:** IC2S2, *SEM, NLP4Democracy, and ALTA.
- **Reviewer:** ACL Rolling Review; Psychological Trauma: Theory, Research, Practice, and Policy; Journal of Biosocial Science; Advances in Methods and Practices in Psychological Science; PLOS Mental Health; Journal of Medical Internet Research.

Teaching, Training, and Student Support

- **Teaching Assistant**, BUSA90543: Natural Language Processing, Melbourne Business School, University of Melbourne, 2025.
- **Sessional Marker**, PSYC90100: Applied Research Methods, Melbourne School of Psychological Sciences, University of Melbourne, 2024.
- Supported Professor Nick Haslam in supervising NLP components of two Psychology Honours projects, including method design, programming, corpus construction and analysis, and results interpretation.
- Trained three research assistants in NLP workflows, corpus construction, and large-scale corpus preprocessing.
- **Academic Tutor**, private tutoring service, 2022–2026. Provided paid tutoring and thesis support to 70+ psychology students across research methods, statistics, academic writing, and thesis preparation.

Media and Outreach

- Nick Haslam & **Naomi Baes**. “[Mental illness, psychiatric disorder or psychological problem. What should we call mental distress?](#)” *The Conversation*, June 2024.
- Nick Haslam, **Naomi Baes**, Ekaterina Vylomova, & Yu Xiao. “[Do we think about anxiety and depression differently now?](#)” *Pursuit*, University of Melbourne, July 2023.
- Melissa A. Wheeler, **Naomi Baes**, Samuel Wilson, & Vlad Demsar. “[What does the common good actually mean?](#)” *The Conversation*, January 2024.